

2024

Bachelor in Business Administration

Course Title: **Database Management System**

Code No: **N/A** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **stamp ordering concurrency control techniques.**14.Illustrate how composite attribute is reduced into relational schema with example?15.Explain constraints of generalization and specialization in ER model.16.What is the importance of database security?Section "C"Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]17.Explain any four types of Database user.18.How normalization helps to design good database? Explain second normal form with example.19.Discuss any four constraints available in database with example.20.Explain the Desirable Properties of Transaction.Section "D"Comprehensive Answer / Case / Situation Analysis Questions:[2 × 10 = 20]21.Draw an ER diagram for Hospital management system with at least five entity set. Implement the concept of mapping cardinality and extended ER feature. Assume suitable attribute sets for each entity set.22.Consider following relational database:Car (reg_no, model, price, color)Person (pid, name, address, phone)owns_by(reg_no, pid, date)Write SQL statement for the following:i) Insert a tuple {1, Ram Bdr, kathmandu} in person relation.ii) Display the model number with highest price.iii) Increase the price of each car by 15%.iv) Find name of person who owns red colored car.v) Display those records of car which model contains at most 4 characters.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. Define Database Language.
2. Write a SQL statement for update operation.
3. What is unary Relationship?
4. List any four types of attributes in ER Model.
5. Define serializable schedule.
6. What is big data?
7. Define deferred data update.
8. What is importance of creating view?
9. State 3NF.
10. Define Roles.

Section B

Short Answer Questions: (Attempt any FIVE Questions)[5 × 3 = 15]

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11. Describe shadow paging concept in brief.
 12. Who is DBA? List the roles of DBA.
 13. Explain the concept of Time stamp ordering concurrency control techniques.
 14. Illustrate how composite attribute is reduced into relational schema with example?
 15. Explain constraints of generalization and specialization in ER model.
 16. What is the importance of database security?

Section C

Long Answer Questions: (Attempt any THREE Questions)[3 × 5 = 15]

17. Explain any four types of Database user.
18. How normalization helps to design good database? Explain second normal form with example.
19. Discuss any four constraints available in database with example.
20. Explain the Desirable Properties of Transaction.

Section D

Comprehensive Answer / Case / Situation Analysis Questions:[2 × 10 = 20]

21. Draw an ER diagram for Hospital management system with at least five entity set. Implement the concept of mapping cardinality and extended ER feature. Assume suitable attribute sets for each entity set.
22. Consider following relational database:
Car (reg_no, model, price, color)
Person (pid, name, address, phone)
owns_by(reg_no, pid, date)
Write SQL statement for the following:
i) Insert a tuple {1, Ram Bdr, kathmandu} in person relation.
ii) Display the model number with highest price.
iii) Increase the price of each car by 15%.
iv) Find name of person who owns red colored car.
v) Display those records of car which model contains at most 4 characters.